Central Forces, Effective Potential, Conic Orbits, and the Runge–Lenz Vector A

Notes prepared by Goutam Sheet for PHY101, 2025

1. Central force basics

A *central force* always points along the radius vector from a fixed center and depends only on the distance r:

$$\mathbf{F}(\mathbf{r}) = F(r)\,\hat{\mathbf{r}}, \qquad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r}.$$

Inverse-square attractions (gravity, unlike charges) take the form

$$\mathbf{F}(r) = -\frac{k}{r^2}\,\hat{\mathbf{r}}, \qquad k > 0.$$

Because $\mathbf{F}$ is strictly radial,

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{0} \quad \Rightarrow \quad \mathbf{L} = m\,\mathbf{r} \times \mathbf{v} \text{ is conserved.}$$

Motion is planar (perpendicular to $\mathbf{L}$). Areal velocity is constant:

$$\frac{1}{2} r^2 \dot{\theta} = \text{const.}$$

Working in polar coordinates (r, θ),

$$v^2 = \dot{r}^{\,2} + (r\dot{\theta})^2, \qquad L = mr^2\dot{\theta} \;\Rightarrow\; \dot{\theta} = \frac{L}{mr^2}.$$

2. Energy and the effective potential

For $V(r) = -k/r$,

$$E = \frac{1}{2}m\dot{r}^{\,2} + \frac{1}{2}m(r\dot{\theta})^2 - \frac{k}{r} = \frac{1}{2}m\dot{r}^{\,2} + \frac{L^2}{2mr^2} - \frac{k}{r}.$$

Define

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{k}{r}, \qquad \frac{1}{2}m\dot{r}^{\,2} + V_{\text{eff}}(r) = E.$$

2.1 Reading $V_{\text{eff}}(r)$ (I am sure you remember the plot from the class

As $r \to 0$, $V_{\text{eff}} \to +\infty$ (centrifugal wall); as $r \to \infty$, $V_{\text{eff}} \to 0^-$. The minimum solves

$$-\frac{L^2}{mr^3} + \frac{k}{r^2} = 0 \;\Rightarrow\; r_0 = \frac{L^2}{mk},$$

with circular-orbit energy

$$E_{\text{circ}} = V_{\text{eff}}(r_0) = -\frac{mk^2}{2L^2}.$$

Orbit classification via the horizontal line E against V_{eff}:

$$\begin{array}{rcl} E = E_{\text{circ}} & \Rightarrow & \text{circle at } r = r_0, \\ E_{\text{circ}} < E < 0 & \Rightarrow & \text{ellipse (bound)}, \\ E = 0 & \Rightarrow & \text{parabola (threshold)}, \\ E > 0 & \Rightarrow & \text{hyperbola (unbound)}. \end{array}$$

Turning points from $E = V_{\text{eff}}(r)$:

$$2mE\,r^2 + 2mk\,r - L^2 = 0 \quad \Rightarrow \quad r_\pm = \frac{-mk \pm \sqrt{m^2k^2 + 2mEL^2}}{2mE}.$$

3. Orbit equation $r(\theta)$: two Newtonian derivations

$$\boxed{r(\theta) = \frac{l}{1 + e\cos(\theta - \theta_0)}}, \qquad \boxed{l = \frac{L^2}{mk}}.$$

3.1 Method A: Energy + the substitution $u = 1/r$

With $\dot{\theta} = \dfrac{L}{mr^2}$ and $\dot{r} = \dfrac{dr}{d\theta}\dfrac{L}{mr^2}$,

$$\left(\frac{dr}{d\theta}\right)^2 \frac{1}{r^2} + 1 - \frac{2kr}{L^2} = \frac{2Er^2}{L^2}.$$

Let $u = 1/r$. Then

$$\left(\frac{du}{d\theta}\right)^2 + u^2 = \frac{2k}{L^2}u + \frac{2E}{L^2} \;\Rightarrow\; \boxed{\frac{d^2u}{d\theta^2} + u = \frac{mk}{L^2}}.$$

Solution:

$$u(\theta) = \frac{mk}{L^2}\Big(1 + e\cos(\theta - \theta_0)\Big) \;\Rightarrow\; r(\theta) = \frac{l}{1 + e\cos(\theta - \theta_0)}.$$

3.2 Method B: Directly from Newton's radial equation

From $m(\ddot{r} - r\dot{\theta}^2) = -k/r^2$ and $\dot{\theta} = L/(mr^2)$,

$$\ddot{r} - \frac{L^2}{m^2r^3} = -\frac{k}{mr^2}.$$

With $u = 1/r$,

$$\dot{r} = -\frac{L}{m}\frac{du}{d\theta}, \quad \ddot{r} = -\frac{L^2}{m^2}u^2\frac{d^2u}{d\theta^2}, \quad r\dot{\theta}^2 = \frac{L^2}{m^2}u^3,$$

giving again

$$\boxed{\frac{d^2u}{d\theta^2} + u = \frac{mk}{L^2}} \;\Rightarrow\; r(\theta) = \frac{l}{1 + e\cos(\theta - \theta_0)}.$$

4. The Runge–Lenz vector A: definition, conservation, and orbit

4.1 Definition

$$\boxed{\mathbf{A} = \mathbf{p} \times \mathbf{L} - mk\,\hat{\mathbf{r}}}, \qquad \mathbf{p} = m\mathbf{v}.$$

It is conserved for the inverse-square law, points to periapsis, and satisfies $|\mathbf{A}| = mk\,e$.

4.2 Proof that $\dfrac{d\mathbf{A}}{dt} = 0$

Differentiate:

$$\frac{d\mathbf{A}}{dt} = \frac{d\mathbf{p}}{dt} \times \mathbf{L} + \mathbf{p} \times \frac{d\mathbf{L}}{dt} - mk\,\frac{d\hat{\mathbf{r}}}{dt}.$$

Since $\dfrac{d\mathbf{L}}{dt} = \mathbf{0}$ for a central force, and $\dot{\mathbf{p}} = -\dfrac{k}{r^2}\hat{\mathbf{r}}$,

$$\frac{d\mathbf{p}}{dt} \times \mathbf{L} = -\frac{k}{r^2}\,\hat{\mathbf{r}} \times \mathbf{L} = -\frac{k}{r^2}\,m\big(\mathbf{r}\,\dot{r} - r\,\mathbf{v}\big).$$

From $\mathbf{r} = r\hat{\mathbf{r}}$,

$$\frac{d\hat{\mathbf{r}}}{dt} = \frac{1}{r}\,\mathbf{v} - \frac{\dot{r}}{r^2}\,\mathbf{r}.$$

Combining terms,

$$\frac{d\mathbf{A}}{dt} = \left[-\frac{mk\dot{r}}{r^2}\,\mathbf{r} + \frac{mk}{r}\,\mathbf{v}\right] - mk\left[\frac{1}{r}\,\mathbf{v} - \frac{\dot{r}}{r^2}\,\mathbf{r}\right] = \mathbf{0}.$$

4.3 Orbit from $\mathbf{A}\cdot\mathbf{r}$

$$\mathbf{A}\cdot\mathbf{r} = (\mathbf{p}\times\mathbf{L})\cdot\mathbf{r} - mk\,\hat{\mathbf{r}}\cdot\mathbf{r} = L^2 - mk\,r.$$

Measure θ from $\mathbf{A}$: $\mathbf{A}\cdot\mathbf{r} = |\mathbf{A}|\,r\cos\theta$. Then

$$|\mathbf{A}|\,r\cos\theta = L^2 - mk\,r \;\Rightarrow\; r = \frac{L^2/(mk)}{1+\frac{|\mathbf{A}|}{mk}\cos\theta} = \frac{l}{1+e\cos\theta},$$

with

$$\boxed{l = \frac{L^2}{mk}}, \qquad \boxed{e = \frac{|\mathbf{A}|}{mk}}.$$

5. Eccentricity–energy relations

5.1 Method I: Conic geometry + l

For an ellipse, $l = a(1-e^2)$, hence $a = \dfrac{L^2}{mk(1-e^2)}$. Energy:

$$E = -\frac{k}{2a} = -\frac{mk^2}{2L^2}(1-e^2) \;\Rightarrow\; \boxed{e^2 = 1 + \frac{2EL^2}{mk^2}}.$$

5.2 Method II: Magnitude of A

Standard identity:

$$\boxed{|\mathbf{A}|^2 = m^2k^2 + 2EL^2} \;\Rightarrow\; \boxed{e = \frac{|\mathbf{A}|}{mk} = \sqrt{1+\frac{2EL^2}{mk^2}}}.$$

6. Orbital types via the effective potential

Circular: $r_0 = \dfrac{L^2}{mk}$, $E_{\text{circ}} = -\dfrac{mk^2}{2L^2}$, $e = 0$.

Ellipses: $E_{\text{circ}} < E < 0$ so $0 < e < 1$; turning radii $r_{\min} = \dfrac{l}{1+e}$ and $r_{\max} = \dfrac{l}{1-e}$.

Parabola: $E = 0$ so $e = 1$ (marginal escape/capture).

Hyperbola: $E > 0$ so $e > 1$ (unbound flyby).

Since $e = \sqrt{1 + \dfrac{2EL^2}{mk^2}}$, at fixed L an increase in E drives

$$\text{circle } (e=0) \ \rightarrow\ \text{ellipse } (0<e<1) \ \rightarrow\ \text{parabola } (e=1) \ \rightarrow\ \text{hyperbola } (e>1).$$

7. Important results (at a glance)

$$\mathbf{F} = -\frac{k}{r^2}\hat{\mathbf{r}}, \quad V(r) = -\frac{k}{r}$$

$$\mathbf{L} = m\,\mathbf{r}\times\mathbf{v} = \text{const}, \qquad \dot{\theta} = \frac{L}{mr^2}$$

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{k}{r}, \quad r_0 = \frac{L^2}{mk}, \quad E_{\text{circ}} = -\frac{mk^2}{2L^2}$$

$$r(\theta) = \frac{l}{1 + e\cos(\theta - \theta_0)}, \qquad l = \frac{L^2}{mk}$$

$$\mathbf{A} = \mathbf{p}\times\mathbf{L} - mk\,\hat{\mathbf{r}}, \qquad \frac{d\mathbf{A}}{dt} = 0, \qquad e = \frac{|\mathbf{A}|}{mk}$$

$$e = \sqrt{1 + \frac{2EL^2}{mk^2}}$$

For in-person questions or to schedule a meeting, email goutam@iisermohali.ac.in. Please indicate whether you prefer a one-on-one conversation or a group discussion.